g) Explain the use of ...

University of Calcutta
M.Sc. 1st Semester Examination, 2025
Computer Science
CSMC103-Advanced Computer Architecture

Marks:30

Time:75 mins

[10x2=20]

1. Answer any ten questions:
   - i. Comment on 'Instruction Execution.'
   - ii. What is memory interleaving, and what are its implications?
   - iii. Illustrate the standard format of IEEE single precision floating point numbers.
   - iv. What are the fundamental assumptions of Von-Neumann architecture?
   - v. What are L1, L2, and L3 caches?
   - vi. Differentiate between a RAM cell and a Cache Memory Cell.
   - vii. Differentiate between RISC and CISC architectures.
   - viii. What are the desirable properties of a processor to be a member of a multiprocessor system?
   - ix. Comment on the impact of branch instructions in pipelined architecture.
   - x. What are the implications of the data routing functions of ICN? Give examples.
   - xi. What is a multi-core processor architecture?
   - xii. Write down the expression for average access time for a three-level cache with usual parameters.

[2x5=10]

2. Answer any two questions:
   - i. Describe the use of normalized mantissa and biased exponent. A 16-bit floating point number has the following representation: the 15^{th} bit is the Sign (s), (9-14) bits store biased exponent (e), and (0-8) bits store normalized mantissa (m). Determine the floating-point value represented by (s, e, m) with 0 as the only special value when 'e' has all 1s.
   - ii. Discuss the use of TLB in mapping virtual addresses to physical addresses.
   - iii. Write an algorithm to test whether or not two input-output pair of desired connection in a MICN will lead to conflict.
   - iv. A computer has a 256 KB, 4-way set associative cache with a block size of 32 bytes. The size of the main memory is 4GB. Determine the number of bits in the tag, set, and offset field in the physical address.